Lean evidence inventory — 2026-10-21

cryptanalysis main `d1a10f6d537dc4e9b5aff02c1e4dcef3e2267c1b`

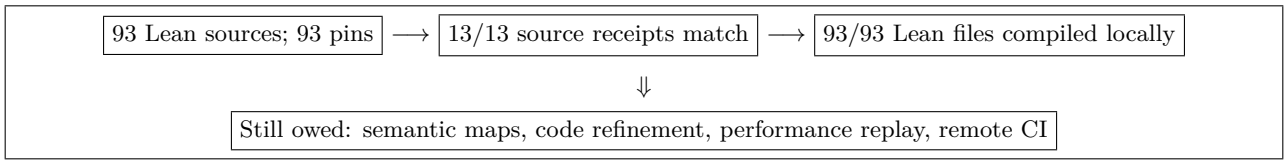
Scope. This change audits tracked Lean source metadata, adds a root toolchain pin, checks source-bound proof receipts, and locally compiles all 93 tracked Lean files in cryptanalysis. It does not semantically review every theorem or experiment across the three requested repositories. No new scientific trial or performance measurement was run.

Delivery. Git smart HTTP returned HTTP 403 ("Your repository is disabled") when pushing this branch. The local commit and evidence remain intact, but this change has no PR or remote CI result yet. The access problem record names the unblock condition.

Inventory and correction

There are 93 tracked Lean files. Before this change, 33 inherited no `lean-toolchain`. The new root pin is `leanprover/lean4:v4.34.1`, while local pins still win. After the change, 92 files resolve to Lean 4.34.1 and the binary-genus-2 module retains Lean 4.24.0.

The new `tools/audit_lean_evidence.py` reads committed files even in sparse checkouts. It checks each pin, scans Lean code outside comments and strings for forbidden proof tokens, and compares dedicated Lean receipt source SHA-256 values and toolchain claims. The observed result was **0 missing pins, 0 files with forbidden tokens, and 13/13 matching source receipts**. CI now runs this static check on relevant changes. A pin, lexical scan, or matching receipt is not a kernel theorem verdict.



Two files have no local `#print axioms`: OrbitPairIndex is covered by its sibling `AxiomAudit.lean`; NearestSelector is imported through `ProofAudit.lean`. The scan reports them for separate review instead of treating file presence as proof. Five parser controls for comments, strings, split tokens, and a real `sorry` passed.

Live proof checks and limits

Check	Observed result and boundary
Tracked Lean corpus	93/93 files compiled on their pinned versions. Source-directed axiom printouts used only propext, Classical.choice and Quot.sound. This is no semantic proof map.
Experiment receipt checks	13/13 scripts passed, covering 28 distinct Lean source paths. Original hardware and timing were not reproduced.
Standalone chains	17 mixed-radix, 24 independent core, and 19 dependent split-prime modules compiled with matching axiom printouts. ProofAudit covered the selector without a local print.
Direct and Lake projects	GaugeTree and OrbitCompression rebuilt; the orbit and Lean 4.24 binary-genus-2 projects built. The latter proves an affine equation under a nonzero denominator, not full Jacobian conformance.
Archived point/model receipts	Six 512-row calibration/holdout checks matched frozen receipts. These are archived point and model comparisons, not fresh performance measurements.
Dedicated receipts	13/13 source hashes match. A matching hash alone is not a theorem verdict.

No corrected scientific measurement arose from this audit. The checked OrbitCompression result itself leaves complete-operation wall time for an isolated host. The existing orbit reference JSON and performance archives were not rewritten; this report's diagram is an evidence visual, not a performance graph. Initial narrow sparse checkouts omitted tracked helpers; after those inputs were included, all 13 receipt scripts passed. The three archived-data checkers were run with their protocol-specified arguments.

Requested scope and status

Requirement	Status
Faithful Lean proofs for every historical experiment across three repos	Partial. All existing cryptanalysis Lean files compiled locally; missing historical lemmas and semantic proof maps remain. The separate autoresearcher workspace verified 62 present targets.
Legitimate code and numerical results	Partial. Six frozen point/model panels replayed; complete implementation refinement and raw performance replay remain.
Graphs and performance dashboards	Partial. Prior autoresearcher PR refreshed nine N13 rows; this change adds an evidence diagram and no new performance number.
Everything good to go	Not established. Code-to-model, empirical, and cross-repository reviews and remote CI remain.

Reproduce the static audit with `python3 tools/audit_lean_evidence.py`; use its `-json` option to save the complete result. The separate `diagram.svg` shows the same proof boundary.